Optimal Market-Neutral Multivariate Pair Trading on the Coffee Market

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March 31, 2026

1 Introduction and Coffee Sector Analysis

The **soft commodities sector**, distinct from hard commodities like metals or energy, comprises agricultural product that are grown rather than extracted. Within that asset class, coffee represents one of the most actively traded commodities globally. The soft group is often referred to as "tropics", because these commodities are grown primarily in tropical or subtropical regions, where in most of the cases information is harder to get. While production largely happens in developing countries, consumptionis mostly concentrated in developed countries.

There are two types of coffee seeds: **Arabica**, that is characterized by higher quality, lower caffeine content and it is mainly traded on the Inter Continental Exchange (ICE) based in New York, while **Robusta** coffee has lower quality, higher caffeine content, it is also more resistant against adverse climatic conditions and is traded in London. The global supply chain is heavily concentrated, with Brazil acting as the dominant producer of Arabica, followed by Colombia, while Vietnam is the leading producer of Robusta. As illustrated in Figure 1, the historical prices of Arabica and Robusta futures clearly reflect these distinct, yet interconnected, market dynamics.

Weather and agronomy are the primary coffee price drivers; for instance, forecasts of frosts or droughts during the flowering phase in key coffee-producing regions like Minas Gerais can cause significant price spikes. These weather shocks typically induce **volatility clustering**. This volatility is further driven by the release of official crop reports. Notably, coffee futures exhibit **asymmetric price transmission** in response to these publications: prices react much more aggressively to bad news (e.g., crop shortages or downward revisions) than to positive news of abundant supply.

Other relevant factors include **logistical issues**, such as container shortages or Red Sea shipping disruptions. This could be reflected in the backwardation of the forward curve. During these periods, near-term contracts trade at a sharp premium to longer-dated months as roasters scramble for immediate physical delivery. Furthermore, rising input costs, particularly for fertilizers, establish a higher breakeven floor for farmers.

Finally, the most important non-weather price driver is the **currency factor**. There is a negative relationship between the US dollar (USD) and currencies of supplier countries, as international coffee contracts and trading are denominated using the former, while payments to exporters/farmers are made in local currencies. Consequently, when a local currency appreciates against the USD, it reduces the payment made in said currency, creating a potential upward pressure on international coffee prices as exporters/farmers try to resist lower returns. For example, the appreciation of the Brazilian real (BRL) against the USD can partly explain some of the recent price bumps.

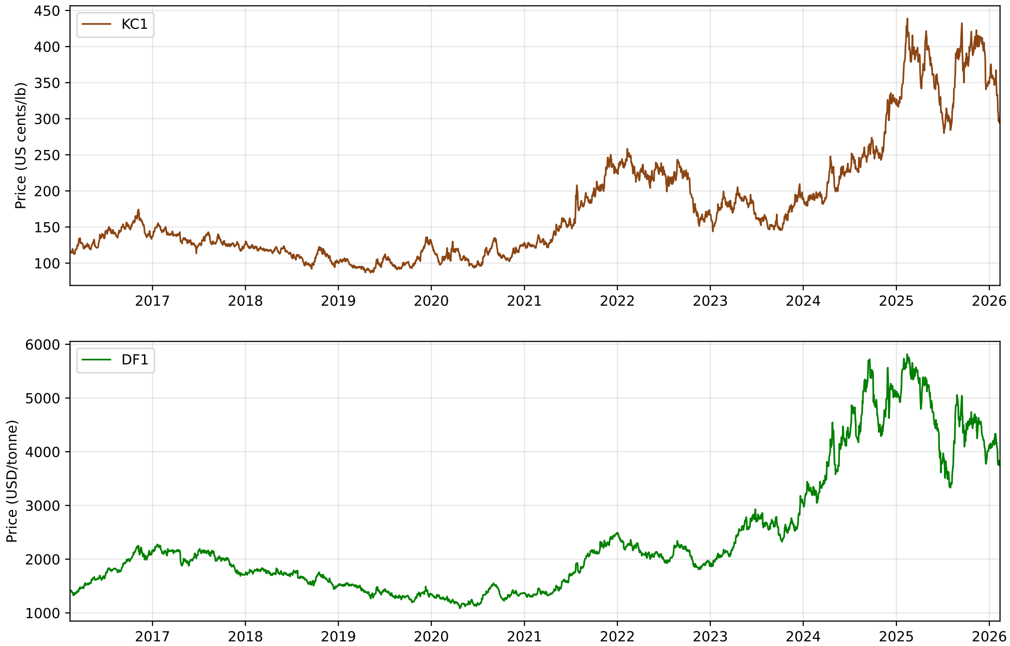

Figure 1: Historical daily prices of KC1 (Arabica) and DF1 (Robusta) coffee futures (2016–2026).

2 Data Collection and Preparation

The data are collected via Bloomberg in OHLCV format and daily frequency, covering the period from 12 February 2016 to 12 February 2026. The asset universe is structured across three levels:

- **Primary anchor:** KC1. It is the Arabica front-month continuous future contract, priced in US cents per pound. It provides the highest liquidity because it always represents the price of the contract closest to expiration in a given moment. The selection of KC1 over Robusta (DF1) as the *primary anchor* is driven by its market dominance — accounting for roughly 65% of global coffee production in 2025 — and its significantly higher trading volume.
- **Sector proxies:** COFF (iPath Coffee ETN), DBA (Invesco DB Agriculture ETF), MOO (VanEck Agribusiness ETF), XLP (State Street Consumer Staples Select Sector ETF), PBJ (Invesco Food & Beverage ETF), DBC (Invesco DB Commodity Index Tracking Fund). These series serve as a benchmark to capture sector dynamics through diversified instruments.
- **Individual universe**: SBUX (Starbucks), BROS (Dutch Bros), KDP (Keurig Dr Pepper), SJM (J.M. Smucker), NSRGY (Nestlé ADR), JDEP (JDE Peet's), MDLZ (Mondelēz), FARM (Farmer Bros.), WEST (Westrock Coffee), QSR (Restaurant Brands), KO (Coca-Cola), LKNCY (Luckin Coffee ADR), MCD (McDonald's), NESN (Nestlé), JVA (Coffee Holding), REBN (Reborn Coffee), THCH (TH International). This represents a highly heterogeneous universe, capturing diversified exposure across the entire coffee value chain, from wholesale roasters to multinational food and beverage conglomerates.

After an initial cleaning, the dataset contains 2,610 rows and 25 close-price columns. Note that not all the assets share the same number of observations and start date: KC1 and most large-cap stocks have the full history, while some companies have a shorter listing history: REBN, WEST, BROS, BRCC, THCH, JDEP and LKNCY were publicly listed only between 2019 and 2022. Therefore, to ensure consistent informational quality these equities are removed from the dataset since they are not matching observational length to the target asset (KC1).

We transform the entire dataset into **log-prices** (natural logarithm of closing prices) rather than working with raw price levels, as the standard convention in quantitative finance. This choice facilitates interpretation and comparison across assets with very different price levels and improves the robustness of statistical tests and parameter estimates.

The 10-year KC1 series shown in Figure 1 directly illustrates why the log-price transformation is appropriate and why unit root tests will be necessary. The series is **clearly non-stationary**: it does not revert to a stable mean, shocks accumulate permanently, and it exhibits very distinct regimes — a prolonged period of prices between 100 and 180 USd/lb from 2016 to 2020, a progressive rise from 2021 driven by global logistics tensions and Brazilian droughts, a historical peak near 400 USd/lb in 2025, then a significant drop mainly due to US tariffs. This profile intuitively confirms the unit root hypothesis that we will test formally.

Before performing any statistical test we define the temporal split between the in-sample (IS) and out-of-sample (OOS) periods. This decision is crucial to avoid any **look-ahead bias**. Therefore, all pairs selection and parameter calibration is done on the IS window and the OOS period data remains untouched until the final performance evaluation. We use a 70%/30% IS-OOS ratio: the **in-sample period** goes from 12/02/2016 to 12/02/2023, yielding 1,826 `KC1` observations, the **out-of-sample period** goes from 13/02/2023 to 12/02/2026, resulting in 783 observations.

Figure 2: Daily log-price series for the `KC1` coffee futures contract. The vertical dashed line divides the in-sample estimation window from the out-of-sample evaluation period.

The foundational requirement for pairs trading is that individual asset price series are **non-stationary**. A stationary series does not admit a long-run equilibrium to revert to, as it is already mean-reverting, implying no exploitable spread dynamics. The **ADF test** results, reported in Table 1, confirm this prerequisite for nearly all the log-price series. With the exception of `QSR` ($p = 0.009$), all candidate equities exhibit p-values well above the 5% significance threshold.

Having established non-stationarity, we test for cointegration between `KC1` and each candidate asset. **Cointegration** is central in pairs trading, as it implies that a linear combination of two $I(1)$ series is stationary, giving rise to a mean-reverting spread. The results indicate that only `SJM` exhibits statistically significant cointegration with `KC1` at the 5% level, allowing us to reject the null of no cointegration. All other equities display weak evidence of cointegration, with p-values between 0.22 and 0.53.

This **lack of strong cointegration** can be explained by different factors. Although many are exposed to coffee through their supply chains, most operate as large and diversified consumer staples companies. Their substantial **pricing power** allows them to reduce input cost fluctuations on to consumers, thereby dampening the direct impact of coffee price movements on equity prices. As a result, their large market capitalizations, diversified revenue streams and exposure to broader macroeconomic factors weaken the long-run statistical relationship with the underlying commodity.

Asset	**N obs**	**Mean**	**Std**	**Skewness**	**Kurtosis**	**Corr**	**ADF**	**E-G**
SJM	1826	4.8040	0.1134	0.1501	−0.7060	0.6502	0.2790	0.0442
KO	1826	3.9094	0.1292	0.4254	−0.9648	0.5525	0.5110	0.2213
MDLZ	1826	3.9303	0.1667	0.1029	−1.4428	0.5280	0.6710	0.2380
JVA	1826	1.4054	0.2572	−0.7746	0.2793	−0.2291	0.4150	0.3607
FARM	1826	2.6682	0.7414	−0.2524	−1.5275	−0.4247	0.9170	0.3721
NSRGY	1826	4.5820	0.1983	−0.0902	−1.4257	0.4182	0.5630	0.4137
NESN	1826	4.5442	0.1762	−0.1654	−1.4273	0.4022	0.6200	0.4167
MCD	1826	5.2235	0.2521	−0.3751	−0.9213	0.3774	0.5860	0.4360
SBUX	1826	4.3128	0.2619	0.2185	−1.3358	0.3366	0.7430	0.4461
QSR	1826	4.0493	0.1557	−1.1256	1.4315	−0.1326	0.0090	0.4584
KDP	1826	3.8232	0.5554	0.5488	−1.4235	0.0983	0.3890	0.5327

Table 1: In-sample summary statistics, correlation with `KC1`, and p-values from Augmented Dickey-Fuller (ADF) and Engle-Granger (E-G) tests.

3 Methodology

After the preliminary stage where the main statistical properties were analyzed to verify the suitability of the series for arbitrage, we proceed by selecting the pairs to be included in the equity basket. Our basket is formed by four pairs (or spreads), i.e. the logarithmic price differences between `KC1` and `SJM`, `KC1` and `KO`, `KC1` and `MDLZ`, `KC1` and `JVA` — henceforth referred to as `KC1/SJM`, `KC1/KO`, `KC1/MDLZ` and `KC1/JVA`, respectively. These assets are selected as they exhibited the highest levels of cointegration with `KC1` during our preliminary screening.

In general, when trying to exploit pairs trading strategies we tipically want pairs that show positive correlation and strong cointegration. In our case, exception made by `KC1/SJM`, other pairs exhibit a **weak cointegration**, as previously pointed out and shown in Table 1. Even if individual cointegration is weak, we rely on the **equity multivariate basket**, using the optimization process to find a "synthetic" relationship that mean-reverts better than single pairs.

Therefore, the pairs trading strategy will rely on the tunable parameters *open_threshold* and *close_threshold*. We calculate the *Z-score* and we open a new position if there is no existing position and the *Z-score* $>$ *open_threshold* $\times$ *standard_deviation*, and we close the position if there is an open position and *Z-score* $<$ *open_threshold* $\times$ *standard_deviation*. We are going to repeat this for the full arbitrage period, i.e. during the OOS window.

Following the procedure of Yang and Malik [5], the thresholds are calibrated using an **IS optimization procedure**. Specifically, we perform a grid search over a range of possible threshold values and evaluate the resulting trading performance using the Sharpe ratio, trying to identify the combination of thresholds that maximizes risk-adjusted returns. A 22-day rolling window is used to estimate the mean and standard deviation of the spread, corresponding to approximately one trading month. Additionally, constraints are imposed during the grid search to prevent excessively narrow entry-exit bands, which could lead to artificially high Sharpe ratios driven by excessive trading frequency rather than genuine mean reversion. This optimization over the IS period yielded relatively wide optimal thresholds: *open_threshold* $= \pm 2.1$, *close_threshold* $= \pm 1.7$. Once a signal is triggered during the OOS window, portfolio weights are solved via the **Gurobi optimizer** (`gurobipy`), which handles the quadratic objective and the constraints given.

As shown in Figure 3, the optimal thresholds lie within a stable region of positive Sharpe ratios rather than an isolated spike, providing reassurance against overfitting. Figure 4 confirms that the Z-score dynamics over the OOS period are consistent with the mean-reverting behaviour assumed by the strategy: all four spreads oscillate within the ± 2.1 band the majority of the time. The `KC1/JVA` spread displays the most irregular pattern, reflecting the lower liquidity of `JVA`.

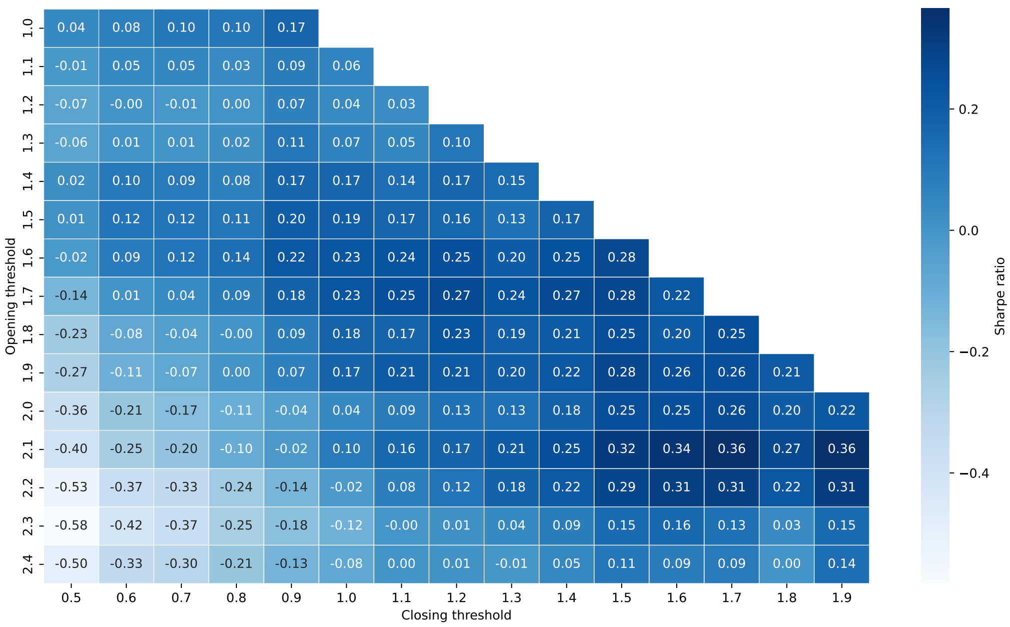

Figure 3: IS Sharpe ratio heatmap from the grid search over *open_threshold* $\in [1, 2.5]$ and *close_threshold* $\in [0.5, 2]$, both defined with a 0.1 step.

To introduce the implementation of the **Beta-neutrality constraint** ($\sum_i \beta_i = 0$), it is essential to distinguish it from the notion market neutrality used in Yang and Malik [5]. In the reference

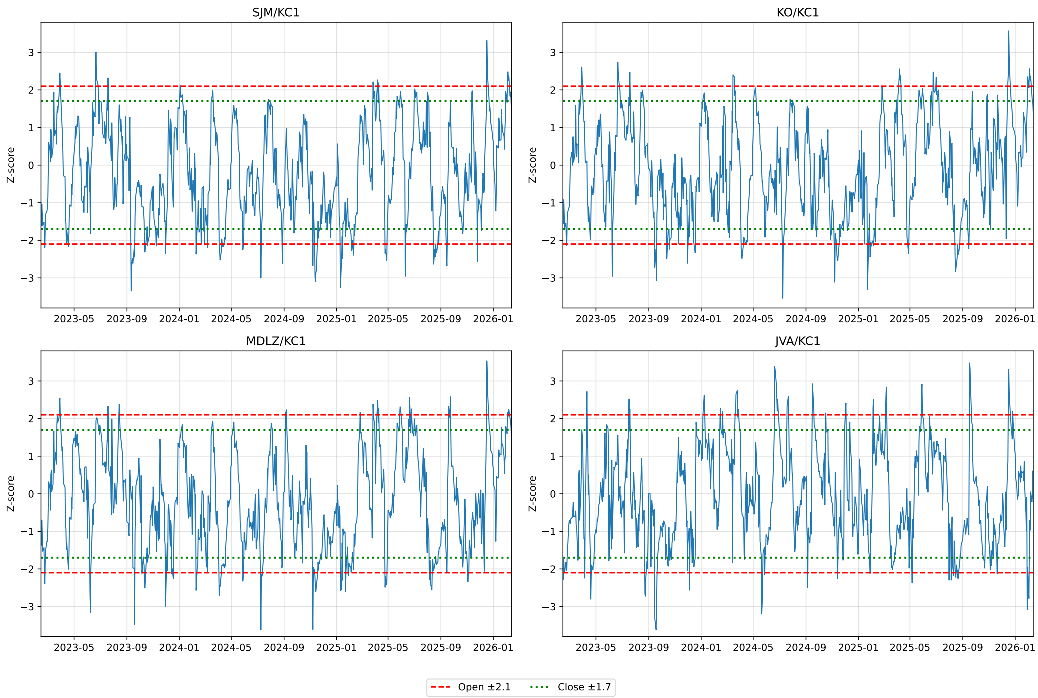

Figure 4: Normalized *Z-score* spreads for the selected equity basket vs. `KC1` during the OOS period. The red and green lines indicate the IS-optimized opening and closing thresholds, respectively.

study, **market neutrality** is achieved by balancing equivalent currency amounts in long and short positions to ensure a stable portfolio value regardless of market direction. This works for highly similar assets like cryptocurrencies expressed in different fiat currencies, where the relationship is near 1-to-1 (correlation ≈ 1 and strong cointegration). In constrast, our strategy involves coffee futures and individual equities, which exhibit varying sensitivities to the underlying commodity: a 1% move in Arabica prices does not result in a 1% move in our stock prices. Therefore, we estimate asset sensitivities via the following **multiple linear regression**.:

$$p_{i,t} = \alpha_i + \beta_{i,\texttt{KC1}}\ p_{\texttt{KC1},t} + \beta_{i,\text{cons\_staples}}\ p_{\text{cons\_staples},t} + \epsilon_{i,t}$$

By utilizing $\beta_{i,\texttt{KC1}}$, i.e. the specific sensitivity of each stock to the *primary anchor*, the optimizer can calculate the weights needed to "immunize" the portfolio against the commodity price swings. The results (Table 2) show that the selected equities are mostly driven by sector dynamics, with $\beta_{i,\texttt{XLP}}$ being significantly different from 0 — exception made again by `SJM`.[1] By controlling for the consumer staples sector, we successfully isolated the residual $\beta_{i,\texttt{KC1}}$ coefficients, which are relatively low, confirming that for large companies, **daily coffee price fluctuations are secondary** to broader sector dynamics. In other words, once we account for how these stocks move with the market, their direct daily sensitivity to the price of Arabica coffee is quite small. Although coffee βs represent a residual portion of the total variance, they still capture a non-negligible source of commodity-specific risk that can be neutralized by explicitly adding the Beta-neutrality constraint.

Asset	α_i	$\beta_{i,\texttt{KC1}}$	$\beta_{i,\texttt{XLP}}$	Adj-R^2
SJM	3.5349 (0.000)	0.2812 (0.000)	−0.0269 (0.182)	0.423
KO	0.2048 (0.000)	−0.0445 (0.000)	0.9958 (0.000)	0.852
MDLZ	−1.0875 (0.000)	−0.1125 (0.000)	1.3571 (0.000)	0.941
JVA	4.6587 (0.000)	0.0618 (0.022)	−0.8666 (0.000)	0.165

Table 2: In-sample estimation results of a multivariate linear regression: coefficients and (p-values) against the *coffee anchor* (`KC1`) and the consumer staples sector proxy (`XLP`).

[1]It should be noted that the coefficients reported in Table 2 are estimated over the full IS period and serve a descriptive purpose. In the OOS backtest, the Beta-neutrality constraint is applied using rolling OLS estimates.

The notation for the variables of interest is adopted from Yang and Malik [5]:

- Let $\boldsymbol{e}$ be a vector of selected equities from coffee's value chain, where $|\boldsymbol{e}| \in \mathbb{Z}^+$ denote the number of stocks.
- Let $\boldsymbol{n}$ be a vector of spreads between the core commodity future and the stock log-prices, where $|\boldsymbol{n}| \in \mathbb{Z}^+$ denote the number of pairs.
- Let $\boldsymbol{t} \in [1, 2, \ldots, T]$ represent the day of a trade execution, where T denotes the total length of the arbitrage (OOS) period.
- Let $\boldsymbol{EP} \in \mathbb{R}^{|\boldsymbol{n}|\times 2}$ be the expected profit matix, recording the expected profit once an arbitrage position is opened. It is calculated as the combination of average returns $r \in \mathbb{R}^{|\boldsymbol{e}|}$ and mean-reversion speed $mr \in \mathbb{R}^{|\boldsymbol{n}|}$ as follows:

$$\boldsymbol{EP} = \begin{bmatrix} mr_{\texttt{KC1/SJM}}\, r_{\texttt{KC1}} & mr_{\texttt{KC1/SJM}}\, r_{\texttt{SJM}} \\ mr_{\texttt{KC1/KO}}\, r_{\texttt{KC1}} & mr_{\texttt{KC1/KO}}\, r_{\texttt{KO}} \\ mr_{\texttt{KC1/MDLZ}}\, r_{\texttt{KC1}} & mr_{\texttt{KC1/MDLZ}}\, r_{\texttt{MDLZ}} \\ mr_{\texttt{KC1/JVA}}\, r_{\texttt{KC1}} & mr_{\texttt{KC1/JVA}}\, r_{\texttt{JVA}} \end{bmatrix}$$

- Let $\boldsymbol{COV} \in \mathbb{R}^{|\boldsymbol{n}|\times 2\times 2}$ be a three-dimensional covariance matrix which contains 2×2 matrices for $|\mathbf{n}|$ pairs. For instance, one the 2×2 matrices is the following:

$$\boldsymbol{\Sigma_n} = \begin{bmatrix} \sigma^2_{\texttt{KC1/SJM}} & \sigma_{\texttt{KC1/SJM,KC1/KO}} \\ \sigma_{\texttt{KC1/KO,KC1/SJM}} & \sigma^2_{\texttt{KC1/KO}} \end{bmatrix}$$

- Let weights $\boldsymbol{W} \in \mathbb{R}^{|\boldsymbol{n}|\times 2}$ be a matrix of weights for every pair, where the first column consists by the long equities with positive weights and the second column is the short equities with negative weights. An example of the matrix is:

$$\boldsymbol{W_n} = \begin{bmatrix} W_{\texttt{KC1}} & W_{\texttt{SJM}} \\ W_{\texttt{KO}} & W_{\texttt{KC1}} \\ W_{\texttt{KC1}} & W_{\texttt{MDLZ}} \\ W_{\texttt{JVA}} & W_{\texttt{KC1}} \end{bmatrix}$$

- Let trading weights $\boldsymbol{tw} \in \mathbb{R}^{|\boldsymbol{e}|}$ be a vector denoting the weight occupation of each equity.
- Let transaction cost $tc \in \mathbb{R}^{\geq 0}$ be a positive constant representing the the percentage deducted from each investment amount as a commission, set to 10 bps in our case.

The **objective of the optimization** is to maximize the expected gain and minimize the λ-weighted risk while respecting three constraints. The **first constraint** is that we can not invest more than what we have, i.e. we can never employ leverage. The **second constraint** ensures that the sum of the current trading weight (long and short weight) does not go over 100% for each individual stock. The **third constraint** is the Beta-neutrality constraint, already discussed above. More formally, we have:

$$\max_{\boldsymbol{W}} \sum_{i=1}^{|\boldsymbol{n}|} \boldsymbol{W}_n \cdot (\boldsymbol{EP}_n \odot [1, -1])' - \lambda \sum_{n=1}^{|\mathbf{n}|} \boldsymbol{W}_n \cdot \boldsymbol{\Sigma}_n \odot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \cdot \boldsymbol{W}'_n$$

$$\text{s.t.} \quad 0 \leq \boldsymbol{W}_{n,\text{long}} \leq 1, \quad -1 \leq \boldsymbol{W}_{n,\text{short}} \leq 0, \quad \forall n \in \mathbf{n}$$

$$tw_e + \sum_{n\in\mathbf{n}} \boldsymbol{W}_{e,\text{long}} - \sum_{n\in\mathbf{n}} \boldsymbol{W}_{e,\text{short}} \leq 1, \quad \forall e \in \mathbf{e}$$

$$\sum_{e\in\mathbf{e}} \beta_{e,\texttt{KC1}} \left(\boldsymbol{W}_{e,\text{long}} + \boldsymbol{W}_{e,\text{short}}\right) = 0$$

where λ is the risk aversion coefficient which impacts the optimization process for how much risk a trader tolerates and $\odot$ denotes the element-wise multiplication (Hadamard product) needed together with the vector $[1, -1]$ and the matrix $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ to ensure that long and short positions contribute with opposite signs to the expected profit.

4 Strategy implementation and results

In this stage, the strategy is initialized with an investment of $10,000 over the three-years OOS period and our goal is to evaluate its performance with respect to other two simple benchmarks: a passive `KC1` buy-and-hold and an equity equally weighted portfolio where the return is defined as

$$r_{p,t} = \frac{r_{\texttt{SJM},t} + r_{\texttt{KO},t} + r_{\texttt{MDLZ},t} + r_{\texttt{JVA},t}}{4}$$

The strategy performance is evaluated using primarily the Sharpe ratio, defined as follows:

$$S_x = \frac{\mathbb{E}[r_x] - r_f}{\sigma_x}$$

where $\mathbb{E}[r_x]$ and σ_x are the annualized log-return and volatility of the x-th strategy and r_f is the risk-free rate set to 4% according to Yang and Malik [5]. The backtest is first conducted using the IS-optimized trading thresholds along with a baseline risk-aversion penalty of $\lambda = 1$. The **Multivariate Pair Trading Strategy** (MPTS) generated an annualized return of 1.48% and a total return of 4.66% over the OOS period, with a Sharpe ratio of -0.165, massively underperforming the passive `KC1` buy-and-hold benchmark which delivered an 18.45% annualized return and a 0.543 Sharpe ratio. In this commodity bull market driven by Brazilian supply shocks and logistical disruptions, the mean-reverting algorithm systematically fades the dominant trend, shorting the outperforming anchor and buying the underperforming equity proxies, acting as a continuous drag on cumulative returns. However, the strategy successfully fulfilled its objective of variance compression. By continuously hedging both the `KC1` directional exposure and the sector-specific component via the Beta-neutrality constraint, the MPTS compressed annualized volatility to 11.17%, representing a significant reduction compared to the 35.69% volatility of the `KC1` benchmark and roughly half of the equally weighted portfolio's 23.02%.

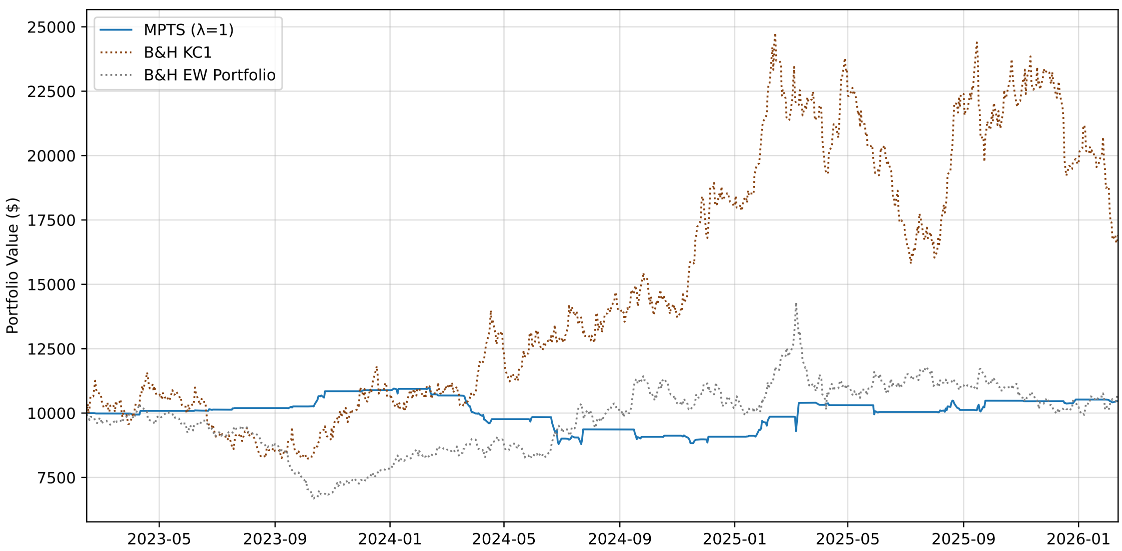

Figure 5: OOS cumulative portfolio value comparison between the multivariate pairs trading strategy (MPTS) with IS-optimized thresholds, a passive buy-and-hold position in the coffee benchmark (`KC1`) and an equally weighted portfolio of selected equities.

4.1 Limitations of IS-optimized signal thresholds

A deeper limitation lies in the IS-optimized thresholds themselves. The optimal thresholds of ±2.1 and ±1.7 calibrated over the 2016–2023 window (a period of **moderate volatility of coffee prices**) reflect a market regime structurally different from the explosive and highly volatile run that characterised the OOS period. As a direct consequence, the *Z-score* rarely breached these bands during the evaluation window, resulting in **very few triggered signals** and a strategy that displayed no trading activity in most of the days. The optimizer's return component, which depends crucially on actual trade frequency to materialize, was effectively neutralized. A heuristic selection of tighter thresholds would lower the entry barrier, activating more signals, increasing trade count, and allowing the expected profit term to contribute meaningfully to the objective, producing a more dynamic and reactive strategy.

To illustrate how threshold selection shapes strategy behaviour, we manually set tighter and different signal thresholds, not as an optimized alternative, but purely as an exploratory exercise to demonstrate the sensitivity of the framework to this parameter. The results of this exercise are presented in the following Figure 6.

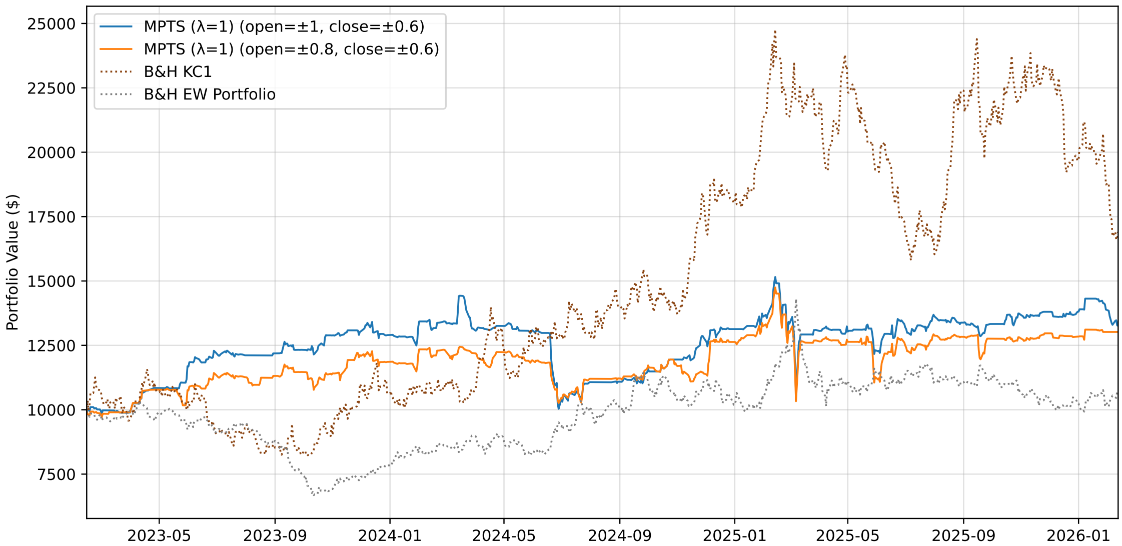

Figure 6: OOS cumulative portfolio value comparison between the MPTS with manually selected thresholds, a passive buy-and-hold in `KC1` and an equally weighted portfolio of selected equities.

With manually selected thresholds of ± 1.0 and ± 0.6, the picture changes substantially: the MPTS generates a Sharpe ratio of 0.349, an annualized return of 9.50% and a total return of 32.57%, with annualized volatility of 20.97%, still well below `KC1`'s 35.69%. Similar results are obtained when tightening the thresholds further to ± 0.8 and ± 0.6, with performance metrics marginally lower, suggesting that the $\pm 1.0/\pm 0.6$ pair represents a more favourable balance between signal frequency and trade quality within this exercise. A closer inspection of Figure 6 reveals a notable pattern: both manually-thresholded strategies perform surprisingly during the first year of the OOS period, accumulating gains steadily before experiencing a sharp drawdown around mid 2024.

Therefore, we deduce that **tighter and lower thresholds activate more signals**, allowing the return component of the optimizer to contribute meaningfully and producing a visibly more dynamic cumulative return profile. However, as previously discussed, these results should be interpreted with caution as the selected thresholds carry an element of look-ahead bias and our purpose here is purely illustrative, i.e. to demonstrate that the poor performance observed under IS-calibrated thresholds reflects a **regime mismatch** rather than a structural flaw in the strategy design.

		KC1	**EWP**	**MPTS$_{(2.1,1.7)}$**	**MPTS$_{(1.0,0.6)}$**	**MPTS$_{(0.8,0.6)}$**
Sharpe	(TC 0.1%)	0.543	0.038	−0.165	0.349	0.321
Sharpe	(TC 0.0%)	0.543	0.038	−0.001	0.528	0.538
Ann. ret.	(TC 0.1%)	18.45%	2.18%	1.48%	9.50%	8.90%
Ann. ret.	(TC 0.0%)	18.45%	2.18%	3.35%	13.75%	14.12%
Total ret.	(TC 0.1%)	69.25%	6.92%	4.66%	32.57%	30.34%
Total ret.	(TC 0.0%)	69.25%	6.92%	10.79%	49.24%	50.74%
Ann. vol.	(TC 0.1%)	35.69%	23.02%	11.17%	20.97%	21.28%
Ann. vol.	(TC 0.0%)	35.69%	23.02%	11.25%	21.11%	21.43%
Max DD	(TC 0.1%)	−36.19%	−35.41%	−19.99%	−30.46%	−29.97%
Max DD	(TC 0.0%)	−36.19%	−35.41%	−19.30%	−29.95%	−29.80%

Table 3: Performance metrics across threshold configurations and transaction cost assumptions. All MPTS results use $\lambda = 1$. Subscripts denote open and close thresholds respectively.

As shown in Table 3, the gap between the **transaction cost** scenarios is non-negligible for the MPTS configurations. The Sharpe ratio for the IS-optimized strategy moves from -0.165 to approximately 0 and for the manual threshold strategies from 0.349 to 0.528 and from 0.321 to 0.538, respectively, confirming that transaction costs do have a visible and quantifiable impact on strategy performance and almost equally the surprising performance of the main benchmark. This sensitivity, while present, remains moderate if compared to the same occurrence in Yang and Malik [5]. This is a direct consequence of the relatively low trading frequency inherent to daily data and the threshold widths employed. By contrast, Yang and Malik [5] operate on intraday data (1-minute and 5-minute intervals) with average holding times of 4-16 hours, where transaction costs accumulate across thousands of trades and represent a proportionally much larger drag on returns. Our strategy, operating at daily frequency with average holding periods of 3.4-7.0 days depending on the threshold configuration, naturally dilutes the per-trade cost burden.

Transaction costs are applied in the implementation of the two passive benchmarks as well, but their effect is by construction negligible and does not affect the comparison. Specifically the `KC1` buy-and-hold is initialized as `(initial_capital - TC * initial_capital)`, deducting a single 10 bps entry cost at the start of the three-year holding period. The equally weighted portfolio applies the same deduction four times (once per equity) yielding an initial cost of 40 bps on the total capital. Given the three-year horizon, these deductions have no measurable effect on the cumulative return trajectory, trivially, as neither benchmark engages any rebalancing or trading over the OOS period.

4.2 Impact of different risk tolerance levels

The **risk-aversion parameter** λ controls the trade-off between expected profit and portfolio variance in the optimization objective. Formally, a higher λ penalizes variance more heavily, pushing the optimizer toward smaller, better-hedged positions at the cost of lower expected returns. The economic interpretation is direct: a trader with $\lambda \to 0$ is a pure return-chaser indifferent to risk, while $\lambda \to \infty$ converges to the **minimum-variance portfolio** (MVP), allocating infinitesimally small weights regardless of the signal. The case $\lambda < 0$, which would correspond to a risk-seeking trader, is excluded from our analysis, as negative risk-aversion is inconsistent with the standard mean-variance optimization framework.

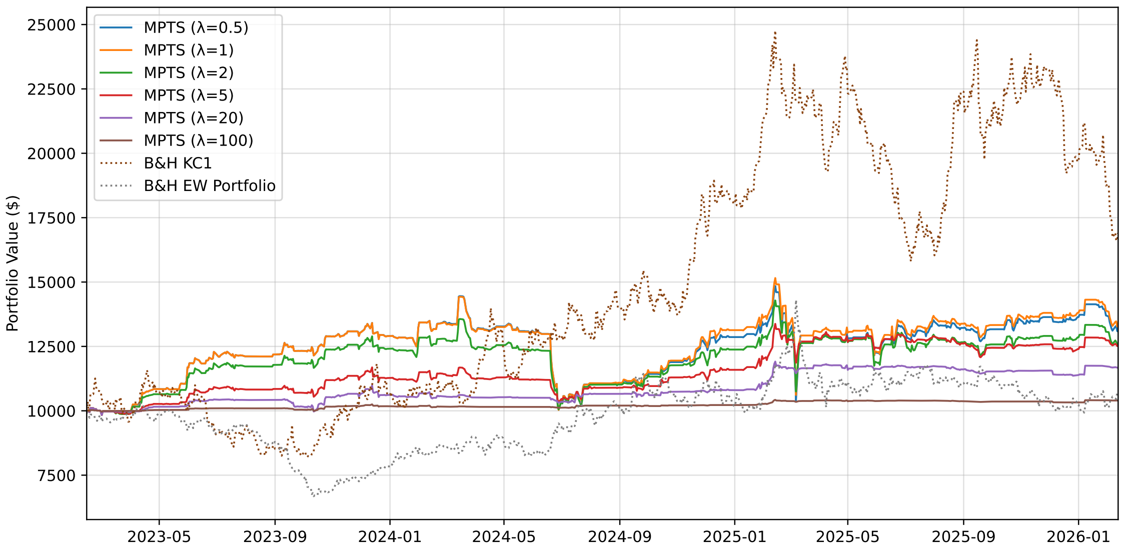

Figure 7: OOS cumulative portfolio value comparison between the MPTS with manually selected thresholds ($\pm 1.0 / \pm 0.6$) for different λ values, a passive buy-and-hold in `KC1` and an equally weighted portfolio of selected equities.

As Table 4 shows, varying λ across the tested range produces surprisingly modest differences in trade-level outcomes. Under manually selected thresholds, the % of winning worsen slightly as λ increases and the average holding period remains stable at 7.0 days throughout, confirming that this parameter governs position sizing and hedging intensity, not entry and exit timing, which is determined solely by the *Z-score* thresholds. The differences are however small enough that no single λ value clearly dominates the others, although $\lambda = 1$ consistently delivers the best overall risk-adjusted performance, as is also visible from the cumulative trajectory in Figure 7. A plausible explanation is that in our setting the expected return signal is weak, so the optimizer's return term provides little discriminatory power across pairs and the risk penalty primarily acts as a regulariser on position size rather than as a selector of superior trades. Under IS-optimized thresholds the picture is essentially the same, with trade counts nearly identical across all λ values, reinforcing that the low signal frequency is a threshold calibration issue rather than a risk-aversion one.

The extreme cases $\lambda = 20$ and $\lambda = 100$ are shown in Figure 7 for pedagogical purposes, illustrating the degenerate behaviour of a highly risk-averse investor. The cumulative portfolio value hugs the initial $10,000 throughout the entire OOS period, as the optimizer systematically allocates near-zero weights to all triggered signals. In the limit, the strategy converges to the minimum-variance portfolio — which, given the Beta-neutrality constraint and the no-leverage restriction, effectively means holding cash.

	IS-optimized thresholds			**Manually selected thresholds**		
Indicator	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$
Number of trades	115	116	116	248	248	258
% of winning	59.1%	58.6%	62.1%	60.9%	62.1%	58.9%
% of losing	40.9%	41.4%	37.9%	39.1%	37.9%	41.1%
% of long win	68.1%	68.1%	69.6%	63.5%	65.0%	61.1%
% of short win	45.7%	44.7%	51.1%	57.7%	58.6%	56.1%
Win/loss ratio	0.84	0.85	0.86	1.09	1.03	1.10
Average loss	−0.0150	−0.0122	−0.0094	−0.0210	−0.0221	−0.0184
Average win	0.0126	0.0103	0.0081	0.0228	0.0227	0.0202
Largest loss	−0.1375	−0.0866	−0.0433	−0.2103	−0.2103	−0.1684
Largest win	0.0971	0.0612	0.0478	0.2126	0.2126	0.1717
Avg holding (days)	3.5	3.4	3.4	7.0	7.0	7.0

Table 4: Strategy indicators across different risk-aversion levels λ for IS-optimized and manually selected thresholds ($\pm 1.0/\pm 0.6$).

Furthermore, Table 4 corroborates the findings discussed in the previous section: the strategy generates only 115-116 trades across three years, compared to 248-258 under manually selected tighter thresholds. This difference is not trivial: a lower trade count means the optimizer has fewer opportunities to express its allocation decisions, and the Beta-neutrality constraint, while correctly formulated, hedges exposures that never materialize into meaningful positions.

5 Conclusion

This paper adapted the multivariate pairs trading framework of Yang and Malik [5] to the coffee market, transposing a methodology originally designed for highly liquid and strongly cointegrated cryptocurrency pairs onto a structurally more complex asset class. The results are instructive precisely because of this transposition.

The optimizer, the Beta-neutrality constraint, and the dynamic allocation mechanism all function as intended: the strategy consistently compresses volatility of the `KC1` benchmark, the Gurobi-based allocation correctly sizes positions across the equity basket, and the rolling β-hedge successfully neutralises the (weak) commodity-specific directional exposure on a daily basis.

The core limitation of this implementation is not methodological but statistical: with three out of four pairs exhibiting Engle-Granger p-values above 0.22, the cointegration foundation of the strategy is necessarily violated. The framework is designed to exploit mean-reversion in spreads, but mean-reversion requires a stable long-run equilibrium and in a commodity market driven by structural supply shocks and currency dynamics, the relationship between a futures contract and consumer staples equities can diverge durably.

Within the constraints of the available universe, however, the strategy demonstrates that the optimizer can extract a positive and controlled return profile even in adverse conditions, which, given the difficulty of the OOS regime, remains an appreciable result.

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